1.3 FORESTRY

Currently, one third of the Earth's land area is covered by forest, which equates to around 4 billion hectares. An individual forest's characteristics and make-up are determined by the latitudinal location and climate. The three distinct types of forest include: boreal, temperate further south, and tropical and subtropical near to the equator.

In order to keep global warming within safe limits, the world needs to develop effective strategies for atmospheric carbon dioxide removal (CDR). Most climate scientists recognise that the only current viable method of CDR at scale is to grow more sustainably managed forests.

How can forestry help CDR?

Trees absorb CO_2 rapidly during growth with the rate declining as the tree reaches maturity. At that point the amount of CO_2 absorbed through photosynthesis is equal to that lost through respiration and decay.

In order to reliably and efficiently use trees and forestry for CDR, a sustainable and long-term use for the timber is required - so that mature trees are felled and new trees are planted. This maintains the overall carbon pool of the forest and yields new material.

Harvesting timber from sustainably managed forests, where replanting maintains or increases timber numbers, increases their capacity as a carbon sink.

Timber buildings as carbon stores

The products made from the felled timber store carbon within their fabric while they are in use.

So using timber as a structural material not only reduces the direct emissions related to construction, reducing the need for carbon intensive materials such as steel and concrete, but the buildings themselves become a new carbon store, with carbon sequestered within their structures.

Ensuring responsible forestry

Certification bodies regulating the legal harvest of timber include The Forest Stewardship Council (FSC) and Programme for the Endorsement of Forest Certification (PEFC).

This third-party certification confirms that forests are managed in a responsible and sustainable way, ensuring diversity, biodiversity, supply, and good conditions for workers.